

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November – December 2011

EE 220 Electrical Machines and Power Circuits

Date: 3.12.2011, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) List the different parts of DC machine and explain any two in details. [07]

(b) Explain the working principle of DC Generator and Motor.

Q - 2 (a) What is the necessity of starter in dc motor? Explain any one in details. [14]

(b) What are the different methods of speed control of dc shunt motor? Explain any one in details.

(c) A 220 V dc machine has an armature resistance of 0.5Ω . If the full load armature current is 20 A. Find the induced emf when the machine acts as (i) generator (ii) motor. Assume the dc machine is to be shunt connected. Neglect the shunt current.

OR

Q - 2 (a) Derive the induced EMF equation of alternator. [14]

(b) Explain synchronous impedance method for measurement of voltage regulation of the alternator.

(c) A 3 phase, 16 pole alternator has a star connected winding with 144 slots and 10 conductors per slot. The flux per pole is 0.03 Weber. Sinusoidally distributed and the speed is 375 rpm. Find the frequency, rpm, phase and line emf. Assume full pitched coil. Take $k_c = 1$.

Q - 3 (a) Derive EMF equation of a single phase transformer. [14]

(b) State and explain the conditions for parallel operation of transformer.

(c) A 2200/250 V Transformer takes 0.5 A at a p.f. 0.3 on open circuit. Find the magnetising and working component of no load primary current.

OR

Q - 3 (a) Why single phase induction motor is not self started? Explain in detail. [14]

(b) Explain the performance of a transformer at no load.

(c) Give the classification of single phase induction motor. Explain any one in detail.

SECTION - II

- Q - 4 (a) Define tariff. State different types of tariff applicable in power system and explain any one in details. [07]
- (b) Explain in details causes of low power factor and disadvantages of the same.
- Q - 5 (a) What is slip in induction motor? State different methods of measurement of slip. [14]
- Explain any one in details.
- (b) Draw & Explain Torque-Slip characteristics of three phase induction motor.

OR

- Q - 5 (a) State and explain importance of power factor improvement. Also state the methods used to improve the power factor. [14]
- (b) Write a comparison of indoor and outdoor substation.
- Q - 6 (a) A Three phase induction motor is wound for 4 poles and is supplied from 50 Hz system. Calculate (i) the synchronous speed (ii) the rotor speed, when slip is 4% (iii) rotor frequency when rotor runs at 600 rpm. [14]
- (b) A consumer has a maximum demand of 200 kW at 40% load factor. If the tariff is Rs. 100 per kW of maximum demand plus 10 paise per kWh, find the overall cost per kWh.

OR

- Q - 6 (a) Give the definition of substation. What is the function of substation? Give the classification of substation. [14]
- (b) Write a technical note on auto transformer.
